Defecography (a dynamic barium enema including lateral views obtained during barium expulsion or a magnetic resonance defecogram) measures changes in rectoanal angle, degree of pelvic floor relaxation and perineal descent, as well as anatomic defects of the rectum such as internal mucosal prolapse, intussusception, and enteroceles or rectoceles. Surgically remediable conditions are identified in only a few patients with severe, whole-thickness intussusception or extremely large rectocele that fills preferentially during attempts at defecation instead of expulsion through the anus. Magnetic resonance defecography provides more information about the structure of the pelvic floor, distal colorectum, and anal sphincters and some information about the function during defecatory effort; however, it is performed in supine rather than seated position.

Neurologic testing (electromyography) is rarely performed in the evaluation of patients with constipation. The absence of neurologic signs in the lower extremities suggests that any documented denervation of the puborectalis results from pelvic (e.g., obstetric) injury or from stretching of the pudendal nerve by chronic, long-standing straining. Constipation is common among patients with spinal cord injuries, neurologic diseases such as Parkinson's disease, multiple sclerosis, and diabetic neuropathy.

Spinal-evoked responses during electrical rectal stimulation or stimulation of external anal sphincter contraction by applying magnetic stimulation over the lumbosacral cord identify patients with limited sacral neuropathies with sufficient residual nerve conduction to attempt biofeedback training.

In summary, a balloon expulsion test is an important component of anorectal manometry as screening tests for anorectal dysfunction. An anatomic evaluation of the rectum or anal sphincters is rarely required when evaluating patients in whom obstructed defecation is suspected. Indications for anatomic evaluation are symptoms of rectal mucosal prolapse, pressure of the posterior wall of the vagina to facilitate defecation (suggestive of anterior rectocele), or prior pelvic surgery that may be complicated by enterocele.

TREATMENT

Constipation

After the cause of constipation is characterized, a treatment decision can be made. Slow-transit constipation requires aggressive medical or surgical treatment; conversely, anismus or pelvic floor dysfunction usually responds to biofeedback management (Fig. 49-5). The remaining ~60% of patients with constipation have normal colonic transit and can be treated symptomatically. Patients with spinal cord injuries or other neurologic disorders require a dedicated bowel regimen that often includes rectal stimulation, enema therapy, and carefully timed laxative therapy.

Patients with constipation are treated with bulk (fiber, psyllium), osmotic (milk of magnesia, lactulose, polyethylene glycol), secretory (linaclotide, lubiprostone, plecanatide, tenapanor), and prokinetic or stimulant laxatives (including diphenyl methanes such as bisacodyl and sodium picosulfate, senna alkaloids, and the 5-HT_4 agonist prucalopride). If a 3- to 6-month trial of medical therapies fails, unassociated with obstructed defecation, the patient should be considered for laparoscopic colectomy with ileorectostomy; however, this should not be undertaken for pain or in the presence of continued evidence of an evacuation disorder or a generalized GI dysmotility. Referral to a specialized center for further tests of colonic motor function is warranted. The decision to resort to surgery is facilitated by the presence of megacolon and megarectum or documented sigmoid or cecal volvulus. The complications after surgery include small-bowel obstruction (11%) and fecal soiling, particularly at night during the first postoperative year. Frequency of defecation is 3–8 per day during the first year, dropping to 1–3 per day from the second year after surgery.

Patients who have a combined (evacuation and transit/motility) disorder should first pursue pelvic floor retraining (biofeedback and muscle relaxation), psychological counseling, and dietetic advice. If symptoms are intractable despite biofeedback and optimized medical therapy, colectomy and ileorectostomy could be considered as long as the evacuation disorder is resolved and optimized medical therapy is unsuccessful. In patients with pelvic floor dysfunction alone, biofeedback training for 3 months has a 70–80% success rate, measured by the acquisition of comfortable stool habits. Attempts to manage pelvic floor dysfunction with operations (internal anal sphincter or puborectalis muscle division) or injections with botulinum toxin have achieved only mediocre success and have been largely abandoned. Persistence of the evacuation disorder despite physical therapy with biofeedback may require regular enemas or consideration of a loop- or end-ileostomy to relieve the intractable constipation.

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50 Unintentional Weight Loss

J. Larry Jameson

Involuntary or unintentional weight loss (UWL) is frequently insidious and can have important implications, often serving as a harbinger of serious underlying disease. Clinically important weight loss is defined as the loss of >5% of body weight over a period of 6–12 months. UWL is not uncommon in individuals aged ≥65 years. It can be challenging to recognize in patients with preexisting obesity or inadequate

documentation of previous weights. There is no identifiable cause in up to one-quarter of patients despite extensive investigation. People with no known cause of weight loss generally have a better prognosis than do those with known causes, particularly when the source is neoplastic. Weight loss in older persons is associated with a variety of deleterious effects, including falls and fractures, pressure ulcers, impaired immune function, and decreased functional status. Not surprisingly, significant weight loss is associated with increased mortality within 1–2 years.

PHYSIOLOGY OF WEIGHT REGULATION WITH AGING

(See also Chaps. 413 and 488) Among healthy aging people, total body weight peaks in the sixth decade of life and generally remains stable until the ninth decade, after which it gradually falls. In contrast, lean body mass (fat-free mass) begins to decline at a rate of 0.3 kg per year in the third decade, and the rate of decline increases further beginning at age 60 in men and age 65 in women. These changes in lean body mass largely reflect the age-dependent decline in growth hormone secretion and, consequently, circulating levels of insulin-like growth factor type I (IGF-I) that occur with normal aging. Loss of sex steroids, at menopause in women and more gradually in men, also contributes to these changes in body composition. In the healthy elderly, an increase in fat tissue balances the loss in lean body mass until very old age, when loss of both fat and skeletal muscle occurs. Age-dependent changes also occur at the cellular level. Telomeres shorten, and body cell mass—the fat-free portion of cells—declines steadily with aging.

Between ages 20 and 80, mean energy intake is reduced by up to 1200 kcal/d in men and 800 kcal/d in women. Decreased hunger is a reflection of reduced physical activity and loss of lean body mass, producing lower demand for calories and food intake. Several important age-associated physiologic changes also predispose elderly persons to weight loss, such as declining chemosensory function (smell and taste), reduced efficiency of chewing, slowed gastric emptying, and alterations in the neuroendocrine axis, including changes in levels of leptin, cholecystokinin, neuropeptide Y, and other hormones and peptides. These changes are associated with early satiety and a decline in both appetite and the hedonistic appreciation of food. Collectively, they contribute to the "anorexia of aging." As noted below, these physiologic changes with aging may be accompanied by social isolation, poverty, and immobility, further contributing to undernutrition.

CAUSES OF UNINTENTIONAL WEIGHT LOSS

Most causes of UWL belong to one of four categories: (1) malignant neoplasms, (2) chronic inflammatory or infectious diseases, (3) metabolic disorders (e.g., hyperthyroidism and diabetes), or (4) psychiatric disorders **(Table 50-1)**. Not infrequently, more than one of these causes can be responsible for UWL. Depending upon patient populations, UWL is caused by malignant disease in a quarter of patients and by organic disease in one-third, with the remainder due to psychiatric disease, medications, or uncertain causes. Risk factors for undiagnosed cancer include a history of smoking, particularly for men, localizing symptoms, and abnormal laboratory tests.

The most common malignant causes of UWL are gastrointestinal, hepatobiliary, hematologic, lung, breast, genitourinary, ovarian, and prostate. Half of all patients with cancer lose some body weight; one-third lose more than 5% of their original body weight, and up to 20% of all cancer deaths are caused directly by cachexia (through immobility and/or cardiac/respiratory failure). The greatest incidence of weight loss is seen among patients with solid tumors. Malignancy that reveals itself through significant weight loss usually has a very poor prognosis.

In addition to malignancies, gastrointestinal diseases are among the most prominent causes of UWL. Peptic ulcer disease, inflammatory bowel disease, dysmotility syndromes, chronic pancreatitis, celiac disease, constipation, and atrophic gastritis are some of the more common entities. Oral and dental problems are easily overlooked and may manifest with halitosis, poor oral hygiene, xerostomia, inability to chew, reduced masticatory force, nonocclusion, temporomandibular joint syndrome, edentulousness, and pain due to caries or abscesses.

TABLE 50-1 Causes of Involuntary Weight Loss

Cancer	**Medications**
Upper gastrointestinal	Sedatives
Lung	Antibiotics
Colon	Nonsteroidal anti-inflammatory drugs
Hepatobiliary	Serotonin reuptake inhibitors
Hematologic	Metformin
Breast	Levodopa
Genitourinary	Angiotensin-converting enzyme inhibitors
Ovarian	Other drugs
Prostate	**Disorders of the mouth and teeth**
Gastrointestinal disorders	Dental caries
Difficulty swallowing	Dysgeusia
Malabsorption	**Age-related factors**
Peptic ulcer	Physiologic changes
Inflammatory bowel disease	Visual impairment
Pancreatitis	Decreased taste and smell
Obstruction/constipation	Functional disabilities
Pernicious anemia	**Neurologic**
Endocrine and metabolic	Stroke
Hyperthyroidism	Parkinson's disease
Diabetes mellitus	Neuromuscular disorders
Pheochromocytoma	Dementia
Adrenal insufficiency	**Social**
Cardiac disorders	Isolation
Chronic ischemia	Poverty
Chronic congestive heart failure	**Psychiatric and behavioral**
Respiratory disorders	Depression
Emphysema	Anxiety
Chronic obstructive pulmonary disease	Paranoia
Renal insufficiency	Bereavement
Rheumatologic disease	Alcoholism
Infections	Eating disorders
HIV	Increased activity or exercise
Tuberculosis	**Idiopathic**
Parasitic infection	
Subacute bacterial endocarditis	

Tuberculosis, fungal diseases, parasites, subacute bacterial endocarditis, and HIV are well-documented causes of UWL. Cardiovascular and pulmonary diseases cause UWL through increased metabolic demand and decreased appetite and caloric intake. Repeated surgeries may lead to weight loss because of reduced caloric intake and increased metabolic demands resulting from a systemic inflammatory response. Uremia produces nausea, anorexia, and vomiting. Connective tissue diseases may increase metabolic demand and disrupt nutritional balance. As the incidence of diabetes mellitus increases with aging, the associated glucosuria can contribute to weight loss. Hyperthyroidism in the elderly may have less prominent sympathomimetic features and may present as "apathetic hyperthyroidism" or T_3 toxicosis **(Chap. 394)**.

Neurologic injuries such as stroke, quadriplegia, and multiple sclerosis may lead to visceral and autonomic dysfunction that can impair caloric intake. Dysphagia from these neurologic insults is a common mechanism. Functional disability that compromises activities of daily living (ADLs) is a common cause of undernutrition in the elderly. Visual impairment from ophthalmic or central nervous system disorders such as a tremor can limit the ability of people to prepare and eat meals. UWL may be one of the earliest manifestations of Alzheimer's dementia.

Isolation and depression are significant causes of UWL that may manifest as an inability to care for oneself, including nutritional needs. A cytokine-mediated inflammatory metabolic cascade can be both a cause of and a manifestation of depression. Bereavement can be a cause

of UWL and is often more pronounced in men. More intense forms of mental illness such as paranoid disorders may lead to delusions about food and cause weight loss. Alcoholism can be a significant source of weight loss and malnutrition.

Elderly persons living in poverty may have to choose whether to purchase food or use the money for other expenses, including medications. Screening questions can probe whether patients have run out of food or whether they routinely purchase less than they need. Institutionalization is an independent risk factor, as up to 30–50% of nursing home patients have inadequate food intake.

Medications can cause anorexia, nausea, vomiting, gastrointestinal distress, diarrhea, dry mouth, and changes in taste. This is particularly an issue in the elderly, many of whom take five or more medications.

ASSESSMENT

The four major manifestations of UWL are (1) anorexia (loss of appetite), (2) sarcopenia (loss of muscle mass), (3) cachexia (a syndrome that combines weight loss, loss of muscle and adipose tissue, anorexia, and weakness), and (4) dehydration. The current obesity epidemic adds complexity, as excess adipose tissue can mask the development of sarcopenia and delay awareness of the development of cachexia. If it is not possible to measure weight directly, a change in clothing size, corroboration of weight loss by a relative or friend, and a numeric estimate of weight loss provided by the patient are suggestive of true weight loss.

Initial assessment includes a comprehensive history and physical, a complete blood count, tests of liver enzyme levels, C-reactive protein, erythrocyte sedimentation rate, renal function studies, thyroid function tests, chest radiography, and an abdominal ultrasound **(Table 50-2)**. Age-, sex-, and risk factor–specific cancer screening tests, such as fecal occult blood, colonoscopy, or mammography, should be performed **(Chap. 75)**. Patients at risk should have HIV testing. All elderly patients with weight loss should undergo screening for dementia and depression by using instruments such as the Mini-Mental State Examination and the Geriatric Depression Scale, respectively **(Chap. 489)**. The Mini Nutritional Assessment (*www.mna-elderly.com*) and the Nutrition Screening Initiative (*http://www.ncbi.nlm.nih.gov/pmc/articles/PMC1694757/*) are also available for the nutritional assessment of elderly patients. Almost all patients with a malignancy and >90% of those with other organic diseases have at least one laboratory abnormality. In patients presenting with substantial UWL, major organic and malignant diseases are unlikely when a baseline evaluation is completely normal. Careful follow-up rather than additional undirected testing is advised because the prognosis of weight loss of undetermined cause is generally favorable.

TABLE 50-2 Assessment and Testing for Involuntary Weight Loss

Indications	Laboratory
5% weight loss in 6 mo	Complete blood count
	Comprehensive electrolyte and metabolic panel, including liver and renal function tests
Body mass index <21	Thyroid function tests
25% of food left uneaten after 7 d	Erythrocyte sedimentation rate
Change in fit of clothing	C-reactive protein
Change in appetite, smell, or taste	Ferritin
Abdominal pain, nausea, vomiting, diarrhea, constipation, dysphagia	HIV testing, if indicated
Assessment	**Radiology**
Complete physical examination, including dental evaluation	Chest x-ray Abdominal ultrasound
Medication review	
Recommended cancer screening	
Mini-Mental State Examination[a]	
Mini-Nutritional Assessment[a]	
Nutrition Screening Initiative[a]	
Simplified Nutritional Assessment Questionnaire[a]	
Observation of eating[a]	
Activities of daily living[a]	
Instrumental activities of daily living[a]	

[a]May be more specific to assess weight loss in the elderly.

TREATMENT

Unintentional Weight Loss

The first priority in managing weight loss is to identify and treat the underlying causes. Treatment of underlying metabolic, social, psychiatric, dental, infectious, or other systemic disorders may be sufficient to restore weight and functional status gradually. Medications that cause nausea or anorexia should be withdrawn or changed, if possible. For those with unexplained UWL, oral nutritional supplements such as high-energy drinks sometimes reverse weight loss. Advising patients to choose appealing foods and to consume supplements between meals rather than with a meal may help minimize appetite suppression and facilitate increased overall intake. Orexigenic, anabolic, and anticytokine agents are not generally recommended. In selected patients, the antidepressant mirtazapine significantly increases body weight, body fat mass, and leptin concentration. However, side effects, including dizziness, fatigue, and somnolence, occur in about 10% of patients. Patients with wasting conditions who can comply with an appropriate exercise program gain muscle protein mass, strength, and endurance and may be more capable of performing ADLs.

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51 Gastrointestinal Bleeding

Loren Laine

Gastrointestinal bleeding (GIB) presents as either overt or occult bleeding. *Overt GIB* is manifested by *hematemesis*, vomitus of red blood or "coffee-grounds" material; *melena*, black, tarry stool; and/or *hematochezia*, passage of red or maroon blood from the rectum. In the absence of overt bleeding, *occult GIB* may present with *symptoms of blood loss or anemia* such as lightheadedness, syncope, angina, or dyspnea; with iron-deficiency anemia; or a positive fecal occult blood test on colorectal cancer screening. GIB is also categorized by the site of bleeding as upper, from the esophagus, stomach, or duodenum; lower, from the colon; small intestinal; or obscure GIB if the source is unclear.

GIB is the most common gastrointestinal condition leading to hospitalization in the United States, accounting for ~530,000 admissions annually and a case fatality of ~2%. Patients generally die from decompensation of other underlying illnesses rather than exsanguination.